

KRISH LALANI

Backend Engineer · Python Developer · Computer Vision Engineer

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PROFESSIONAL SUMMARY

Computer Engineering graduate (CGPA 9.03 / 10) building production backend systems and computer vision pipelines end-to-end. Across three engineering roles, shipped FastAPI and Django services scaling to 500+ users, dual-model YOLOv8 inference on ESP32-CAM edge hardware, and real-time IoT dashboards over MQTT, ThingsBoard, and Grafana. Comfortable across the full stack — from REST API architecture and PostgreSQL schema design to GPIO debugging on Jetson Orin Nano and Raspberry Pi. Ships systems that work in the field, not just in notebooks.

CORE SKILLS

Languages	Python, JavaScript, SQL, HTML5, CSS3
Backend	Django, FastAPI, Flask, Node.js, Express.js, REST APIs, JWT, RBAC
CV & AI	YOLOv8 (Ultralytics), YOLOv5, OpenCV, Object Detection, Image Stitching, MJPEG Streaming
Edge & IoT	NVIDIA Jetson Orin Nano, Raspberry Pi (GPIO), ESP32 / ESP32-CAM, Arducam CSI/UVC, MQTT
Databases	PostgreSQL, MySQL, MongoDB, Firebase, Supabase
DevOps	Git, Azure DevOps, Docker, Linux (Ubuntu), CI/CD, Postman
Monitoring	Grafana, ThingsBoard IoT, Real-time Dashboards, Sensor Integration
Web & Data	BeautifulSoup, Selenium, EJS, JSON, HTTP / REST

EXPERIENCE

Dev Industry	Apr 2026 – Present
Junior Software Developer	Hybrid
- Architecting vision-based detection systems and scalable REST API backends for client deployments, owning design through deployment on Azure DevOps. - Driving full SDLC execution — architecture, code review, and CI/CD pipelines — using Git-based workflows across cross-functional teams.	
Empire Circuits LLC	Sep 2025 – Apr 2026
Python Developer	New Jersey, USA (Remote)
- Engineered PondGuard — a dual-model YOLOv8 inference pipeline processing live ESP32-CAM MJPEG streams via Flask, triggering relay-controlled sprinklers, strobes, and buzzers at sub-second latency. - Built an OpenCV multi-frame image-stitching QA pipeline for PCB inspection; statistical defect overlays cut manual review time and improved detection consistency. - Integrated ThingsBoard IoT + Grafana for real-time sensor telemetry, exposing field-deployed metrics through interactive dashboards backed by MQTT.	
Infotact Solution	Apr 2025 – Jul 2025
Python Developer	Bengaluru, Karnataka

- Built Droplify — an AI-powered price tracker monitoring 10+ e-commerce platforms in real time using FastAPI, MongoDB, and async scraping (BeautifulSoup / Selenium).
- Integrated 5+ third-party product data APIs and shipped trend-analytics dashboards driving an automated price-drop alert engine.

CHARUSAT University — In-house Internship

Jun 2024 — Aug 2024

Backend Developer & Team Lead

Anand, Gujarat

- Led a 4-engineer team to deliver placeStar — a placement and exam platform serving 500+ students, with JWT authentication and role-based access control.
- Architected the Node.js / Express + PostgreSQL REST API; ran sprint planning, schema design reviews, and query optimization across the team.

SELECTED PROJECTS

PondGuard — Autonomous Bird Detection & Deterrent System · [live demo](#)

Dual YOLOv8 inference (YOLOv5su + custom-trained model) running on a live ESP32-CAM stream; Flask orchestrates relay-triggered sprinkler, strobe, and buzzer responses, with Firebase push alerts and a browser-accessible live MJPEG feed.

Python · YOLOv11 · Flask · OpenCV · MQTT · Firebase · ESP32-CAM

CarePlus — Healthcare Management Platform · [live demo](#)

Role-based dashboards for Doctors, Staff, and Patients with real-time Supabase sync and a hardened Django REST backend handling appointments, prescriptions, and patient records.

Python · Django · PostgreSQL · Supabase · REST APIs

AMC Connect — Annual Maintenance Contract Portal · [live demo](#)

B2B portal managing AMC agreements end-to-end — service schedules, technician dispatch, and automated renewal reminders — built for operational teams.

Node.js · Express.js · PostgreSQL · Supabase · REST APIs

Droplify — AI-Powered E-Commerce Price Tracker · [live demo](#)

Real-time price monitoring across 10+ platforms with an async scraping pipeline, ML-based price-drop predictions, and an automated alert engine that surfaces actionable signals.

Python · FastAPI · MongoDB · BeautifulSoup · Selenium

placeStar — Campus Placement & Exam Management · [live demo](#)

Centralized platform for 500+ students covering placement drives, exam scheduling, and student records; JWT / RBAC authentication and an optimized PostgreSQL schema reduce latency on common access paths.

Node.js · Express.js · PostgreSQL · JWT · EJS

AutoStitch QA Analyzer · [live demo](#)

Multi-frame image stitching pipeline on Raspberry Pi using OpenCV and NumPy; statistical defect overlays automate PCB visual inspection and reduce manual QA time.

Python · OpenCV · NumPy · Raspberry Pi

EDUCATION

CHARUSAT University, Anand

2023 — Present

B.E. in Computer Engineering · CGPA: 9.03 / 10.00

A.V. Parekh Technical Institute, Rajkot

2020 — 2023

Diploma in Computer Engineering · CGPA: 9.89 / 10.00 · Top 10 Merit Award